Emberz: plan

Fireworks hackathon, November 2026. Numan and Matt. Built on Fireworks plus open models; no fallbacks for now.

1. What we're building

Emberz gives your kid, your grandma or you a guarded phone number. Every call to it passes through our line. A small model fine-tuned on Fireworks reads the live transcript and stays silent on normal calls. It speaks up when a call turns into a scam, when a stranger asks a kid for personal info, or when the caller is an AI voice.

- **The protected person** hears a quiet tip that the caller can't hear.
- **The caretaker** (mom, an adult child) gets an alert call. They can listen live, join the call with one tap, and get a summary afterwards.
- **The caretaker writes the rules** in plain words, such as "never share the door code" or "tell me about any talk of money".
- **The real number is never given out**, so it stays out of data brokers' lists, leaks and strangers' hands.

What we prove: - **Main claim:** a small model fine-tuned on Fireworks using only generated calls catches more risky calls before the ask (money, codes, address, meeting up) than Llama Guard 3, gpt-oss-safeguard and a frontier model. It raises fewer false alarms on normal calls, is faster, and costs much less per call-minute. - **Second result, built in parallel:** AI-voice detection on real phone audio, compared with an off-the-shelf detector and Telnyx's deepfake check.

Why it's new (checked 2026-11-02): Google and Samsung detect scams on the phone itself and warn only the person on the call. OnGuardAI texts family a red flag, with no transcript, no AI-voice check, no caretaker rules and no kids. Hiya and Truecaller check the voice only. Kids' phones check texts, not calls. No one streams the call to a caretaker under rules the caretaker writes, and no one publishes an open model with a reproducible eval.

2. The stack (decided)

Part	Choice
Phone line	Telnyx on EmberDial's account, isolated: a **new** Call Control app and **new** numbers. Never touch EmberDial's apps, connections, numbers or webhooks
Public URL for webhooks	`cloudflared` tunnel to the line server on Numan's Mac
Speech-to-text	**NVIDIA Nemotron Speech Streaming 0.6B** (open, NVIDIA Open Model License), run locally with **sherpa-onnx** INT8. Streams in 160 ms chunks, one stream per side
Training-data generator	**Nemotron 3 Ultra** on Fireworks serverless (Batch API for bulk)
Held-out test generator	**GLM 5.3 Flash** on Fireworks serverless: a different model family, so the test isn't the generator grading itself
Guardian model	**Qwen3-4B-Instruct-2507**, LoRA SFT on Fireworks, served on an on-demand deployment
Baselines	Keyword rules, Llama Guard 3 8B (on-demand), gpt-oss-safeguard-20b (on-demand), untuned Qwen3-4B (same deployment), **Kimi K3** serverless
Dev stand-in for the guardian	A cheap serverless model with the same prompt and schema, while the line is being built
AI-voice detector	Local: frozen wav2vec2 front end + small head, trained on public data. Telnyx `deepfake_detection` ($0.01 per call) as the commercial baseline
AI test callers	Our own agents on Vapi and Retell (free credits) calling our number

Part	Choice
Text-to-speech, for transcription noise in training data	Kokoro (Apache-2.0), local

Fireworks checks done (2026-11-02): - The key works. - Nemotron 3 Ultra answers on serverless. - Qwen3-4B and Qwen3-8B support LoRA. - Nemotron Nano 3 does not support LoRA on Fireworks, so it can't be the guardian. - Fireworks' hosted speech-to-text is unavailable on our key (503 / not found), so speech-to-text runs locally on open Nemotron.

3. How a call works

```
caller dials the guarded number (Telnyx)
  |
  v
answer (+ deepfake_detection) -> notice: "This call is protected by Emberz
  |                                        and may be transcribed"
  |-- media stream, both tracks, one WebSocket --> line server (Numan's Mac)
  |      |-- Nemotron streaming ASR, one stream per side ("Caller" / "Grandma")
  |      |-- AI-voice detector (first 2-4 s) + reply-gap timing
  |      |-- guardian model on Fireworks: rules + last turns -> JSON
  |      '-- rule layer: smoothing, cooldowns, allowed contacts -> action
  v
conference with the protected person's real phone
  |-- coach: whisper only the protected person hears
  |-- caretaker alert call: "Press 1 to listen" -> listen, barge in
  |-- live console (web): transcript, risk lines, alerts
  '-- hang-up: summary to the caretaker
```

4. Contracts: the glue that lets us work in parallel

All events are JSON over a WebSocket or HTTP in the line server. The schemas live in `contracts/` and are agreed in hour 0, so nobody waits on anybody.

`turn`: Line to Guardian, one final transcript segment.

```
{"call_id": "c_123", "turn": 7, "speaker": "caller|protected", "text": "can you read me the
 ↪  code",
 "t_start_ms": 18320, "t_end_ms": 19950, "gap_ms": 1400}
```

`guardian_request`: rules, the turn window and signals.

```
{"call_id": "c_123", "profile": "elder|kid|adult",
 "rules": ["Never share codes or passwords", "Alert me about any money talk", "Allowed:
  ↪  Grandpa (+1555...)"],
 "caller_known": false, "ai_voice": {"score": 0.91, "source": "detector|telnyx|timing"},
 "turns": [{"turn": 1, "speaker": "caller", "text": "..."}]}
```

`guardian_response`: one per scored turn (section 6).

`voice_signal`: Voice to Line.

```
{"call_id": "c_123", "at_ms": 2400, "ai_score": 0.91, "model": "w2v-head-v1|telnyx|timing"}
```

`console_event`: Line to the console: `call_started`, `turn`, `risk`, `alert`, `coach`, `caretaker_joined`, `summary`, `call_ended`.

`dataset_row`: Model, one generated call.

```
{"call_id": "s_000123", "split": "train|dev|test_heldout", "generator":
 ↪  "nemotron-3-ultra|glm-5.3-flash",
 "profile": "kid", "rules": ["..."], "rule_flip_of": null, "scam_type": "impersonation_bank",
```

```
"turns": [{"turn": 1, "speaker": "caller", "text": "...", "label": {"risk": "none",
 ↪  "category": "none", "stage": "hook"}}],
"ask_turn": 9, "benign_twin": false}
```

results_row: one row per model, test set and metric, used for the charts.

```
{"model": "guardian-v1", "test_set": "heldout|bothbosu|roleplay|pan12", "metric":
 ↪  "catch_before_ask", "value": 0.83, "n": 300, "ci95": [0.79, 0.87]}
```

5. Work packages

Big chunks, each with an owner, outputs, what unblocks it, and when it's done. **M** = Model (Matt), **L** = Line (Numan), **V** = Voice (Numan), **D** = Demo (both).

M1. Data factory (Matt)

- **Builds:** the scenario grid, which is protected person (kid 8-12, teen, elder, adult) x caller type (scams by family, stranger, groomer at the tactics level, real bank, doctor's office, school, family member, friend, AI appointment bot) x rule set x persona.
- **Self-play:** Nemotron 3 Ultra plays the caller from a hidden plan (stages, the exact ask turn, tactics) and another call plays the protected person. Labels come from the plan.
- **Extras:**
  - Benign twins (35%): normal calls that reuse scam words.
  - Rule-flip pairs (10%): the same words under different rules.
  - Disclosure calls: the protected person gives away personal info.
  - One scam type held out entirely.
  - Transcription noise on part of the set: Kokoro, then 8 kHz, then Nemotron ASR.
- **Hard rule: grooming stays at the level of tactics. No sexual content involving minors is ever generated.** A conversation stops at the request.
- **Held-out test set:** made by GLM 5.3 Flash with different persona seeds.
- **Outputs:** `data/train.jsonl`, `data/dev.jsonl`, `data/test_heldout.jsonl` (gitignored), dataset card, overlap check.
- **Unblocked by:** nothing. Starts in hour 0.
- **Done when:** about 3,000 training calls and 500 held-out calls exist, with turn-level labels, and pass validation against the schema.

M2. Guardian fine-tune and deployment (Matt)

- **Builds:** the training-set builder (calls into windowed chat samples with JSON targets), SFT v1, one multi-LoRA deployment (base + v1 + v2), and SFT v2 after the adversary round.
- **Budget:** cap v1 at about 24M billed tokens. Keep every ask, near-ask and benign-twin window, and subsample easy negatives. **Always pass an explicit LoRA rank (8-16).** A missing rank triggers full training at twice the price.
- **Outputs:** model IDs, deployment ID, `guardian/client.py` (the `guardian_request` to `guardian_response` call).
- **Unblocked by:** M1's first 500 calls. A tiny test job first confirms how multi-turn samples are billed.
- **Done when:** v1 answers on the deployment with valid JSON and beats the untuned base (B3) on the dev set.

M3. Eval harness, baselines and charts (Matt)

- **Builds:** scorers for every metric in section 7, baselines B0-B4, the real test sets (BothBosu, PAN-12 subset, our role-play calls from V2), and the charts.
- **Run early:** B0 and B4 on BothBosu plus 50 dev calls in the first hours. Labels for the real sets are committed to git before any model runs on them.
- **Outputs:** `results/*.jsonl` in the `results_row` format, `eval/charts/*.png`.
- **Done when:** one command re-runs every model on every test set and redraws the charts.

L1. Guarded line (Numan)

- **Builds:**
  - The new Telnyx Call Control app and 3 new numbers: Grandma, kid, caretaker test.
  - Webhook signature checks and the `cloudflared` tunnel.
  - Answer with `deepfake_detection` on, play the notice, stream both tracks to the WebSocket.
  - Conference with the protected person's phone.
- **Checks first:** conference whisper and barge (`supervisor_role`), and whether the original caller ID shows on the protected phone.
- **Done when:** a call to the guarded number rings a real phone, both sides stream to the server, and the Telnyx deepfake result arrives by webhook.

L2. Live brain (Numan)

- **Builds:**
  - Nemotron streaming ASR per side (sherpa-onnx), with -law decode, stateful resampling and turn segmentation.
  - The `turn` events and the guardian client (a dev stand-in model until M2 lands).
  - The rule layer: smoothing, cooldowns, allowed contacts, and AI voice + unknown caller + ask = escalate.
  - The whisper coach (text-to-speech heard only by the protected person).
  - The caretaker alert call ("Press 1 to listen"), joining as a listener, and barge-in.
- **Done when:** a role-played scam call produces a whisper and an alert call within about 1.5 s of the trigger line.

L3. Caretaker console and summaries (Numan, or Matt if free)

- **Builds:** a phone-friendly web page with the live transcript of both sides, risk lines (AI voice, scam, disclosure, grooming), alerts, a join button and the end-of-call summary (one serverless call at hang-up).
- **Unblocked by:** a fake `console_event` stream.
- **Done when:** the full demo script (section 8) runs on screen.

V1. AI-voice detector (Numan)

- **Builds:** frozen wav2vec2 features over the first 2-4 s of caller audio, plus a small logistic head. Also timing and text cues: reply gap (people ~0.2 s, voice agents ~1.3-2.4 s), "I'm an AI assistant", missing "um"s.
- **Training data:**
  - ASVspoof 5, which includes 8 kHz codec conditions.
  - In-the-Wild.
  - Kokoro renders of our scripts.
  - Human audio over the same Telnyx path.
  - **Not ElevenLabs output:** its use policy bans training on it.
- **Baselines:** off-the-shelf W2V2-AASIST (MIT) and Telnyx `deepfake_detection`.
- **Done when:** EER and AUC on our phone set (from V2) for audio only, timing only and combined, with n and intervals.

V2. Real test calls (Numan records, Matt labels blind)

- **Builds:**
  - 5 of our own AI agents on Vapi and Retell, each with a different voice, making about 10 calls each to our number.
  - About 50 human calls over the same path.
  - About 40 role-play guardian calls between teammates and consenting friends: scam, disclosure, grooming tactics, AI caller and benign hard negatives, with scripts written blind.
- **Done when:** the audio, transcripts and labels are committed (labels before any model runs).

D1. Demo, README and submission (both)

- **Builds:** the rehearsed demo (section 8), the README with results, charts and legal notes, the 60-second video, the Fireworks endpoint and the repo.

- **Done when:** submitted.

Sync points

1. **Hour 0, 30 minutes:** agree the contracts, the metrics and the JSON schema. Then split.
2. **Hour 6:** M1's first data feeds M2. L2 switches to real events.
3. **Hour 10:** guardian v1 on the deployment. L2 points at it for the first live test.
4. **Hour 14:** V2 recordings and labels handed to M3 and V1.
5. **Hour 18:** feature freeze, then rehearsal, video and results.

Critical path

M1, then M2 (v1), then M3 (scores), then D1. The line (L1 to L3) and voice (V1, V2) run alongside it.

6. The guardian model

Output per scored turn:

```
{"risk": "none|watch|alert|urgent",
 "category": "scam.impersonation_bank_gov | scam.tech_support | scam.family_emergency |
  ↪ scam.prize | scam.romance_investment | disclosure.location_school |
  ↪ disclosure.credentials_otp | disclosure.financial_id | disclosure.home_alone |
  ↪ grooming.secrecy | grooming.isolation | grooming.gifts_flattery | grooming.photo_request
  ↪ | grooming.meeting_request | vulgar | threat_extortion | self_harm | none",
 "severity": 0, "source": "caller|protected_person", "stage": "hook|trust|pressure|ask",
 "evidence": [{"turn": 7, "quote": "exact words from the transcript"}],
 "action": "none|log|coach|notify_caretaker|suggest_hangup|urgent_alert",
 "coach_line": "15 words max", "caretaker_note": "20 words max"}
```

- **Headline categories:** scam, disclosure and grooming tactics.
- **Second slice, scored separately:** vulgar language, threats and self-harm. Llama Guard is strong on these, so the slice is reported on its own.
- **An AI voice alone is at most `watch`.** AI voice plus an unknown caller plus an ask escalates. A pharmacy reminder bot lights the badge but sends no alert. The FCC treats AI voices as "artificial" under the TCPA (FCC 24-17) and requires such calls to identify who is calling (47 CFR 64.1200(b)).
- **Rules vary across training**, so the model follows the rules it's given instead of memorizing one set.

7. Scoring

Metric	Definition
Catch-before-ask (headline)	% of risky calls at or above `alert` before the labeled ask turn, reported at 5 or fewer false alarms per 100 benign calls
False alarms	Alerts per 100 benign calls, hard negatives included
Lead time	Turns between the first alert and the ask
Rule-flip accuracy	% of rule-flip pairs decided per the rules
Unseen scam type	Recall on the scam type held out of training
Category F1	Macro-F1 over categories
Evidence match	% of quotes found exactly in the transcript
Speed	p50 and p95 per scored turn, in milliseconds
Cost	$ per call-minute, including deployment GPU time

	Baseline	Where
B0	Keywords + personal-info regex	Local

	Baseline	Where
B1	Llama Guard 3 8B, last turn vs a 6-turn window	Fireworks on-demand
B2	gpt-oss-safeguard-20b with the caretaker's written rules	Fireworks on-demand
B3	Qwen3-4B untuned, same prompt	Same deployment as ours
B4	Kimi K3 zero-shot	Fireworks serverless

Test sets, reported separately, each with n and a 95% interval: 1. The held-out generated set. 2. BothBosu: 20 real scam calls. 3. A PAN-12 subset: grooming in chat, not phone calls, and we say so. 4. Our ~40 role-play calls.

Charts: - Catch-before-ask vs false alarms, one point per model. - Speed vs quality. - A per-category breakdown. - v1 vs v2. - Separately, AI-voice detection.

8. The demo

The screen shows the live console. Three phones: Grandma's, the kid's and Mom's.

1. **An AI voice scams Grandma.** Our AI caller dials as "her bank". About 2 s in, the console shows **AI voice**. On "read me the code", Grandma hears: "Banks never ask for codes. Say you'll call the number on your card." Mom's phone rings with the alert.
2. **Mom joins.** She presses 1 and listens, then barges in.
3. **A stranger calls the kid.** "Parents home? What street?" sends Mom the alert and the live transcript.
4. **Same words, different rules.** Grandpa, on the allowed list, asks the same thing and it stays silent.
5. **An AI that isn't a scam.** A pharmacy reminder bot calls. The AI badge lights, no alert.
6. **Close.** The overlay shows "alerted N turns before the ask", then the chart.

A judge can call the guarded number from their own phone and try to fool it.

9. Fireworks budget ($100: two $50 accounts, which can't be pooled)

Estimates use $8 per hour for an H100 or H200 (billed per second; start-up isn't billed) and $0.50 per million SFT tokens for models up to 16B.

Account	Line item	Estimate
A (Matt)	SFT v1, about 24M tokens	~$12
A (Matt)	SFT v2, about 8M tokens	~$4
A (Matt)	Guardian deployment, eval windows (B3 + v1 + v2 back to back), about 2 h	~$16
A (Matt)	Guardian deployment, rehearsal and demo, about 1.5 h	~$12
A (Matt)	Reserve: never spend the last $6, or the demo deployment stops	$6
B (Numan)	Generation through the Batch API (50% off): train and held-out sets	~$10
B (Numan)	B4 Kimi K3, plus summaries and the dev stand-in	~$8
B (Numan)	B1 Llama Guard + B2 safeguard deployments, about 1.5 h each	~$24
	Total	**~$92**

Rules: - Deployments run only during eval windows and the demo. Create them with `--min-replica-count 0 --scale-to-zero-window 5m`, and delete them right after. - Run `firectl deployment list` on both accounts at

the end of every session. - Turn Auto Reload off and set a $50 spend limit on each account. - Spend dashboards lag, so keep a ledger in `results/ledger.md` of GPU start and stop times and token counts. - Install `firectl` in hour 0 and check deployment shapes.

10. Consent (for the README)

- **Every call starts with a notice.** It covers all-party-consent states such as California (Penal Code 632.7).
- **The protected person opts in.** Federal law needs one party's consent. A caretaker can't consent for a competent adult.
- **Kids:** a line the parent owns. Under the amended COPPA rule (enforced since April 22, 2026), a child's voice is personal information, so we need parental consent plus written retention and security policies.
- **Privacy by default:** alerts and summaries first, with the live transcript opt-in or opened only after an alert. No voiceprints, and data deleted after 7 days.
- **Hackathon data:** generated calls, public datasets within their licenses, and calls between consenting teammates and friends. **No EmberDial customer data, ever.**

11. Repo layout

```
PLAN.md, README.md
contracts/   JSON schemas for every event and row (section 4)
factory/     M1 scenario grid, self-play, labels, splits
guardian/    M2 training-set builder, SFT and deploy scripts, client
eval/        M3 scorers, baselines, charts
line/        L1 + L2 Telnyx app, media stream, ASR, rule layer, coach, alerts
console/     L3 caretaker console
voice/       V1 AI-voice detector
recordings/  V2 (gitignored audio; labels committed)
results/     results rows, ledger
data/        gitignored
```

12. Cut order (if time runs short)

1. The adversary round (v2).
2. B1 Llama Guard.
3. Barge-in.
4. The second category slice.

Never cut: B2, B3 and B4 on the held-out set, and the benign-twin false-alarm number. They are the proof.

Sources (checked 2026-11-02)

- Fireworks: SFT · tunable models · deploying LoRAs · pricing · Llama Guard 3 8B · gpt-oss-safeguard-20b
- Speech-to-text: Nemotron Speech Streaming 0.6B · sherpa-onnx Nemotron
- Telnyx: media streaming · deepfake detection
- Landscape: Google Scam Detection on Galaxy S26 · OnGuardAI · Aura Call Protection · Hiya AI voice detection
- Data and research: BothBosu scam datasets · Shen et al. 2024 · PAN-12 · ASVspoof 5 · In-the-Wild · W2V2-AASIST · Kokoro-82M · ElevenLabs use policy
- Law: 18 USC 2511 · CA Penal Code 632.7 · FCC AI voices under TCPA · COPPA amendments

Not legal advice: get a lawyer's read before launching a real product.